# THERMODYNAMICS

## 16.1 Internal energy

Candidates should be able to:

1. understand that internal energy is determined by the state of the system and that it can be expressed as the sum of a random distribution of kinetic and potential energies associated with the molecules of a system
2. relate a rise in temperature of an object to an increase in its internal energy

## 16.2 The first law of thermodynamics

Candidates should be able to:

1. recall and use $W = p\Delta V$ for the work done when the volume of a gas changes at constant pressure and understand the difference between the work done by the gas and the work done on the gas
2. recall and use the first law of thermodynamics $\Delta U = q + W$ expressed in terms of the increase in internal energy, the heating of the system (energy transferred to the system by heating) and the work done on the system

**1. *O/N/09/42/Q2***

**(a)** State what is meant by the *internal energy* of a gas.

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**(b)** The first law of thermodynamics may be represented by the equation

$$\Delta U = q + w.$$

State what is meant by each of the following symbols.

$+\Delta U$ ..............................................................................................................................

$+q$ ...................................................................................................................................

$+w$ ...................................................................................................................................
[3]

**(c)** An amount of 0.18 mol of an ideal gas is held in an insulated cylinder fitted with a piston, as shown in Fig. 2.1.

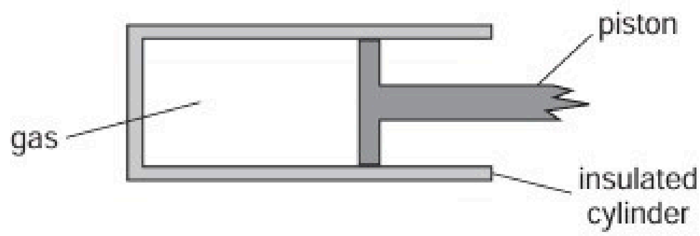


**Fig. 2.1**

Atmospheric pressure is $1.0 \times 10^5$ Pa.

The volume of the gas is suddenly increased from $1.8 \times 10^3\,cm^3$ to $2.1 \times 10^3\,cm^3$.

For the expansion of the gas,

**(i)** calculate the work done by the gas and hence show that the internal energy changes by 30 J,

[3]

**2. O/N/10/41/Q2**

**(a)** **(i)** State the basic assumption of the kinetic theory of gases that leads to the conclusion that the potential energy between the atoms of an ideal gas is zero.

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**(ii)** State what is meant by the *internal energy* of a substance.

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**(iii)** Explain why an increase in internal energy of an ideal gas is directly related to a rise in temperature of the gas.

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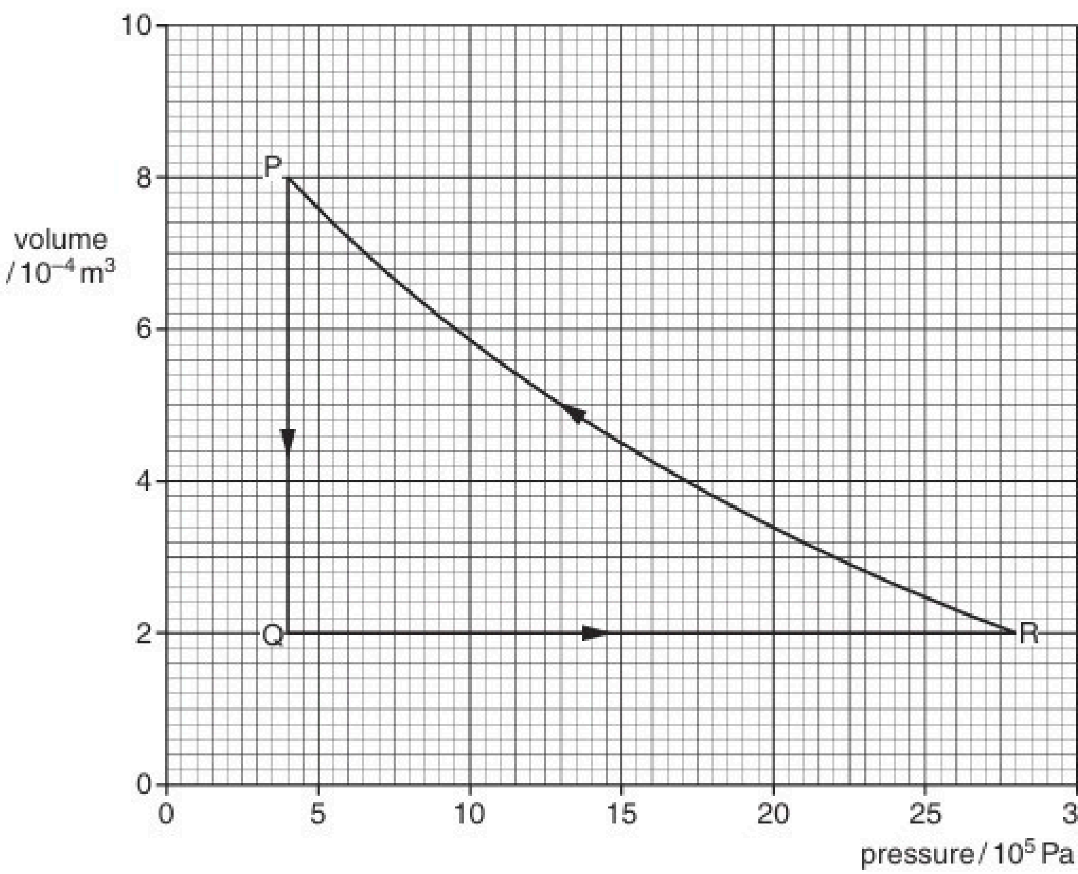
**(b)** A fixed mass of an ideal gas undergoes a cycle PQRP of changes as shown in Fig. 2.1.



**(i)** State the change in internal energy of the gas during one complete cycle PQRP.

change = ............................................. J [1]

**(ii)** Calculate the work done on the gas during the change from P to Q.

work done = ............................................... J [2]

**(iii)** Some energy changes during the cycle PQRP are shown in Fig. 2.2.

| change | work done on gas / J | heating supplied to gas / J | increase in internal energy / J |
|---|---|---|---|
| P $\rightarrow$ Q | ............................ | –600 | ............................ |
| Q $\rightarrow$ R | 0 | +720 | ............................ |
| R $\rightarrow$ P | ............................ | +480 | ............................ |

**Fig. 2.2**

Complete Fig. 2.2 to show all of the energy changes. [3]

**3. M/J/11/42/Q4**

**(a)** The first law of thermodynamics may be expressed in the form

$$\Delta U = q + w.$$

Explain the symbols in this expression.

$+\Delta U$ ..........................................................................................................................

$+q$ ..........................................................................................................................

$+w$ ..........................................................................................................................
[3]

**(b) (i)** State what is meant by *specific latent heat*.

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**(ii)** Use the first law of thermodynamics to explain why the specific latent heat of vaporisation is greater than the specific latent heat of fusion for a particular substance.

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***4. M/J/12/42/Q3***

**(a)** State what is meant by the *internal energy* of a system.

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**(b)** State and explain qualitatively the change, if any, in the internal energy of the following systems:

**(i)** a lump of ice at 0 °C melts to form liquid water at 0 °C,

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**(ii)** a cylinder containing gas at constant volume is in sunlight so that its temperature rises from 25 °C to 35 °C.

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***5. O/N/12/41/Q2***

A student suggests that, when an ideal gas is heated from 100 °C to 200 °C, the internal energy of the gas is doubled.

**(a)** **(i)** State what is meant by *internal energy*.

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**(ii)** By reference to one of the assumptions of the kinetic theory of gases and your answer in **(i)**, deduce what is meant by the internal energy of an ideal gas.

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**(b)** State and explain whether the student's suggestion is correct.

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**6. O/N/13/43/Q2**

**(a)** **(i)** State what is meant by the *internal energy* of a system.

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**(ii)** Explain why, for an ideal gas, the internal energy is equal to the total kinetic energy of the molecules of the gas.

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**(b)** The mean kinetic energy $<E_K>$ of a molecule of an ideal gas is given by the expression

$$<E_K> = \frac{3}{2}kT$$

where $k$ is the Boltzmann constant and $T$ is the thermodynamic temperature of the gas.

A cylinder contains 1.0 mol of an ideal gas. The gas is heated so that its temperature changes from 280 K to 460 K.

**(i)** Calculate the change in total kinetic energy of the gas molecules.

change in energy = ............................................. J [2]

**(ii)** During the heating, the gas expands, doing $1.5 \times 10^3$ J of work.
State the first law of thermodynamics. Use the law and your answer in **(i)** to determine the total energy supplied to the gas.

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total energy = ................................................ J [3]

**7. M/J/14/41/Q3**

The volume of 1.00 kg of water in the liquid state at 100 °C is $1.00 \times 10^{-3}\,m^3$. The volume of 1.00 kg of water vapour at 100 °C and atmospheric pressure $1.01 \times 10^5$ Pa is $1.69\,m^3$.

**(a)** Show that the work done against the atmosphere when 1.00 kg of liquid water becomes water vapour is $1.71 \times 10^5$ J.

[2]

**(b)** **(i)** The first law of thermodynamics may be given by the expression

$$\Delta U = +q + w$$

where $\Delta U$ is the increase in internal energy of the system.

State what is meant by

**1.** $+q$,

.............................................................................................................................. [1]

**2.** $+w$.

.............................................................................................................................. [1]

**(ii)** The specific latent heat of vaporisation of water at 100 °C is $2.26 \times 10^6\,J\,kg^{-1}$.

A mass of 1.00 kg of liquid water becomes water vapour at 100 °C.

Determine, using your answer in **(a)**, the increase in internal energy of this mass of water during vaporisation.

increase in internal energy = ...................................................... J [2]

***8. M/J/15/41/Q2***

**(a)** State what is meant by *internal energy*.

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**(b)** The variation with volume $V$ of the pressure $p$ of an ideal gas as it undergoes a cycle ABCA of changes is shown in Fig. 2.1.

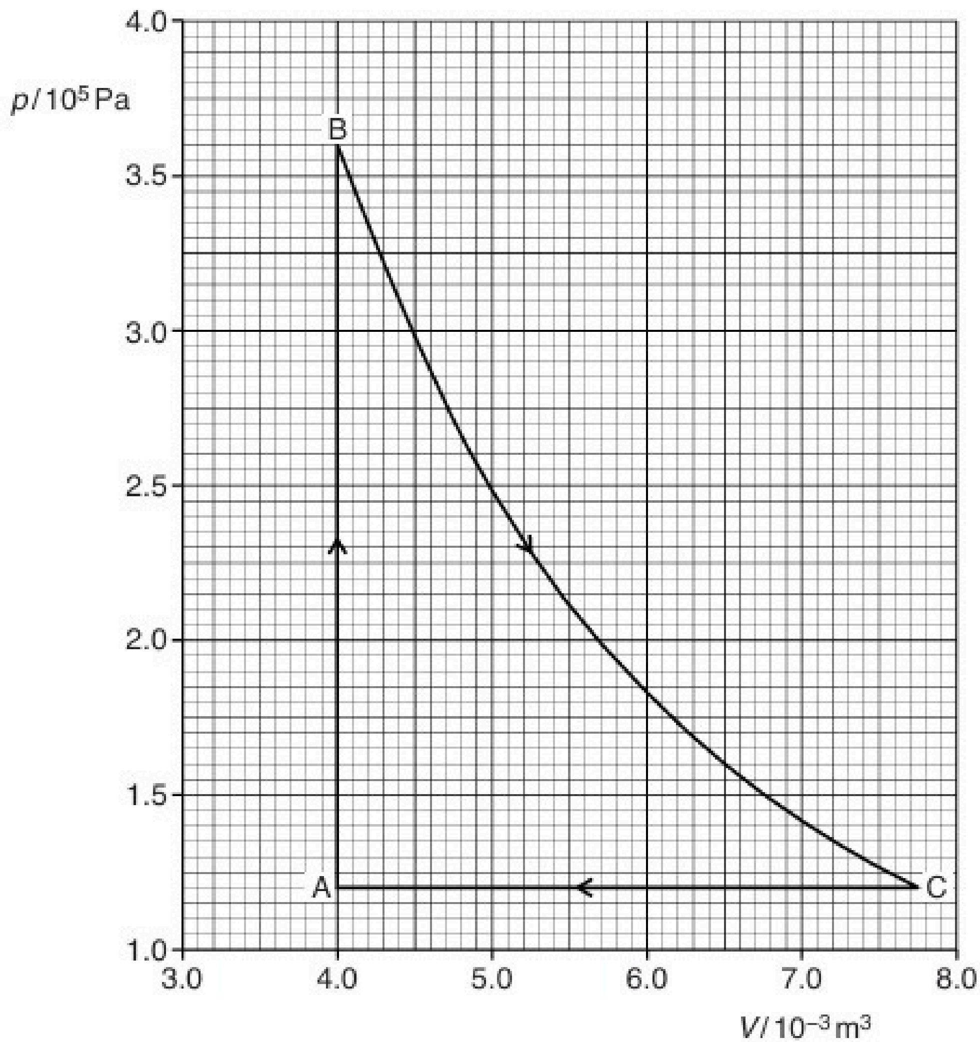


**Fig. 2.1**

The temperature of the gas at A is 290 K. The temperature at B is 870 K.

Determine

**(i)** the amount, in mol, of gas,

amount = .................................................. mol [2]

**(ii)** the temperature of the gas at C.

temperature = ...................................................... K [2]

**(c)** Explain why the change from C to A involves external work and a change in internal energy.

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## *9. O/N/15/41/Q3*

**(a)** State an expression, in terms of work done and heating, that is used to calculate the increase in internal energy of a system.

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**(b)** State and explain, in terms of your expression in **(a)**, the change, if any, in the internal energy

**(i)** of the water in an ice cube when the ice melts, at atmospheric pressure, to form a liquid without any change of temperature,

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**(ii)** of the gas in a tyre when the tyre bursts so that the gas suddenly increases in volume. Assume that the gas is ideal.

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### 10. O/N/16/41/Q2

An ideal gas initially has pressure $1.0 \times 10^5$ Pa, volume $4.0 \times 10^{-4}$ m$^3$ and temperature 300 K, as illustrated in Fig. 2.1.

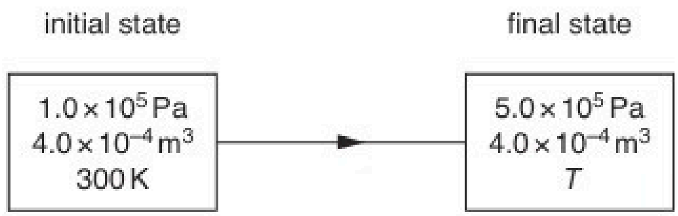


**Fig. 2.1**

A change in energy of the gas of 240 J results in an increase of pressure to a final value of $5.0 \times 10^5$ Pa at constant volume.
The thermodynamic temperature becomes $T$.

**(a)** Calculate

**(i)** the temperature $T$,

$T$ = ........................................................ K [2]

**(ii)** the amount of gas.

amount = .................................................... mol [2]

**(b)** The increase in internal energy $\Delta U$ of a system may be represented by the expression

$$\Delta U = q + w.$$

**(i)** State what is meant by the symbol

**1.** $+q$,

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**2.** $+w$.

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[2]

**(ii)** State, for the gas in **(a)**, the value of

**1.** $\Delta U$,

$\Delta U$ = ............................................................ J

**2.** $+q$,

$+q$ = ............................................................ J

**3.** $+w$.

$+w$ = ............................................................ J
[3]

[Total: 9]

***11. O/N/16/42/Q3***

**(a)** State what is meant by the *internal energy* of a system.

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**(b)** Explain, by reference to work done and heating, whether the internal energy of the following increases, decreases or remains constant:

**(i)** the gas in a toy balloon when the balloon bursts suddenly,

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**(ii)** ice melting at constant temperature and at atmospheric pressure to form water that is more dense than the ice.

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[Total: 8]

## 12. F/M/17/42/Q2

**(a)** The first law of thermodynamics can be represented by the expression

$$\Delta U = q + w.$$

State what is meant by the symbols in the expression.

$+\Delta U$ ..............................................................................................................................

$+q$ ..............................................................................................................................

$+w$ ..............................................................................................................................

[2]

**(b)** A fixed mass of an ideal gas undergoes a cycle ABCA of changes, as shown in Fig. 2.1.

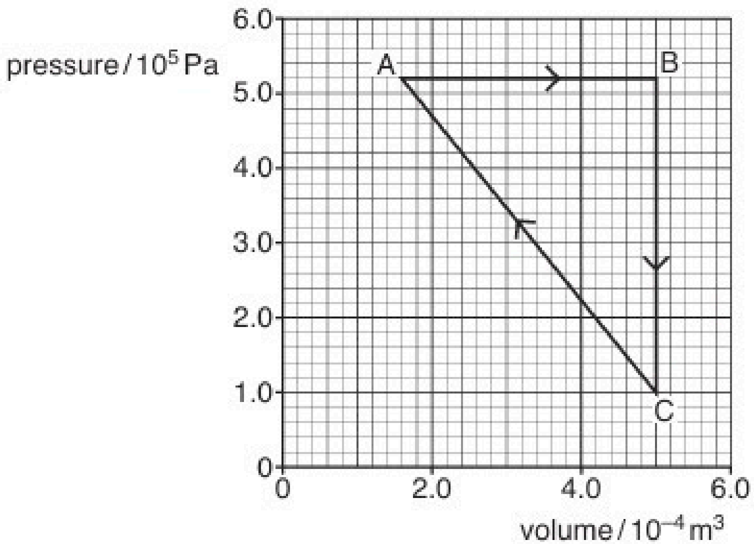


**Fig. 2.1**

**(i)** During the change from A to B, the energy supplied to the gas by heating is 442 J.

Use the first law of thermodynamics to show that the internal energy of the gas increases by 265 J.

[2]

(ii) During the change from B to C, the internal energy of the gas decreases by 313 J.

By considering molecular energy, state and explain qualitatively the change, if any, in the temperature of the gas.

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(iii) For the change from C to A, use the data in **(b)(i)** and **(b)(ii)** to calculate the change in internal energy.

change in internal energy = .......................................................... J [1]

(iv) The temperature of the gas at point A is 227 °C. Calculate the number of molecules in the fixed mass of the gas.

number = .............................................................[2]

[Total: 10]

## *13. M/J/18/41/Q3*

**(a)** **(i)** State what is meant by the *internal energy* of a system.

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.................................................................................................................................[2]

**(ii)** Explain why, for an ideal gas, the change in internal energy is directly proportional to the change in thermodynamic temperature of the gas.

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**(b)** A cylinder of volume $1.8 \times 10^4\,cm^3$ contains helium gas at pressure $6.4 \times 10^6\,Pa$ and temperature 25 °C.
Helium gas may be considered to be an ideal gas consisting of single atoms.

Calculate the number of helium atoms in the cylinder.

number = ..............................................................[3]

[Total: 8]

**14. O/N/18/41/Q2**

**(a)** State what is meant by the *internal energy* of a system.

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**(b)** An ideal gas undergoes a cycle of changes as shown in Fig. 2.1.

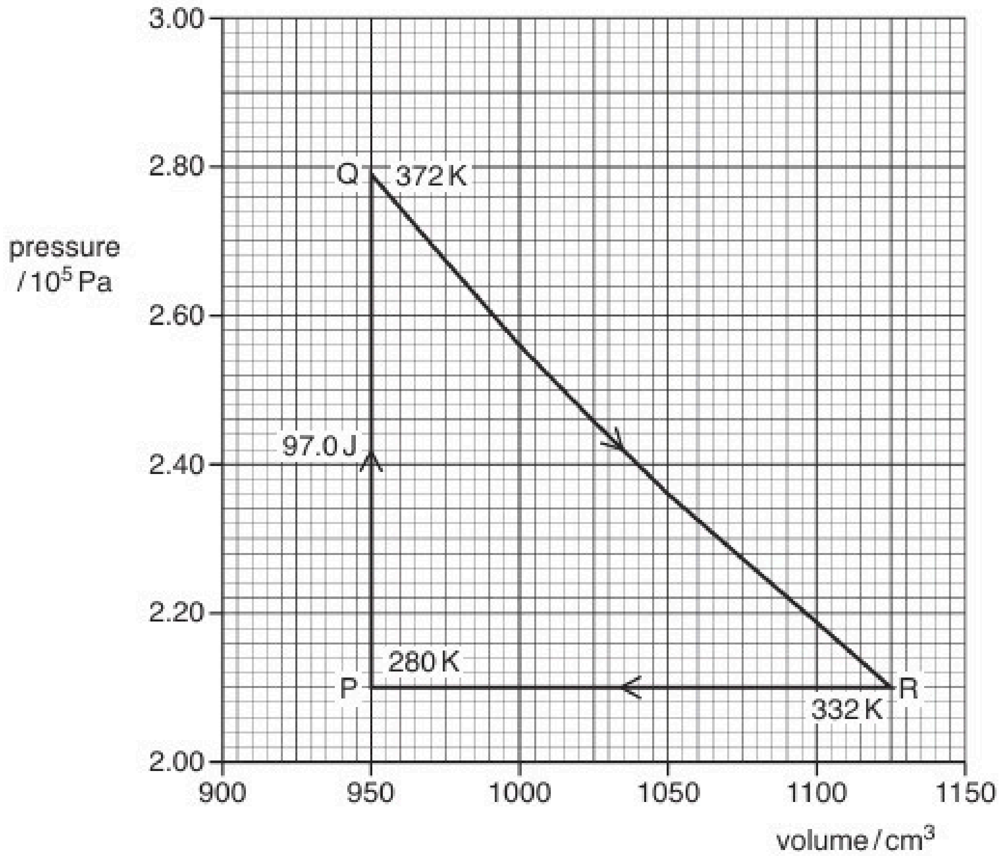


**Fig. 2.1**

At point P, the gas has volume 950 $cm^3$, pressure $2.10 \times 10^5$ Pa and temperature 280 K.

The gas is heated at constant volume and 97.0 J of thermal energy is transferred to the gas. Its pressure and temperature change so that the gas is at point Q on Fig. 2.1.

The gas then undergoes the change from point Q to point R and then from point R back to point P, as shown on Fig. 2.1.

Some energy changes that take place during the cycle PQRP are shown in Fig. 2.2.

| | change P → Q | change Q → R | change R → P |
|---|---|---|---|
| thermal energy transferred to gas / J | +97.0 | 0 | ........................ |
| work done on gas / J | ........................ | −42.5 | +37.0 |
| increase in internal energy of gas / J | ........................ | ........................ | ........................ |

**Fig. 2.2**

**(i)** State the total change in internal energy of the gas during the complete cycle PQRP. Explain your answer.

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**(ii)** On Fig. 2.2, complete the energy changes for the gas during

**1.** the change P → Q,

**2.** the change Q → R,

**3.** the change R → P.

[5]

[Total: 9]

***15. M/J/19/42/Q2***

**(a)** The first law of thermodynamics may be expressed in the form

$$\Delta U = q + w.$$

**(i)** State, for a system, what is meant by:

**1.** $+q$

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**2.** $+w$.

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[2]

**(ii)** State what is represented by a negative value of $\Delta U$.

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**(b)** An ideal gas, sealed in a container, undergoes the cycle of changes shown in Fig. 2.1.

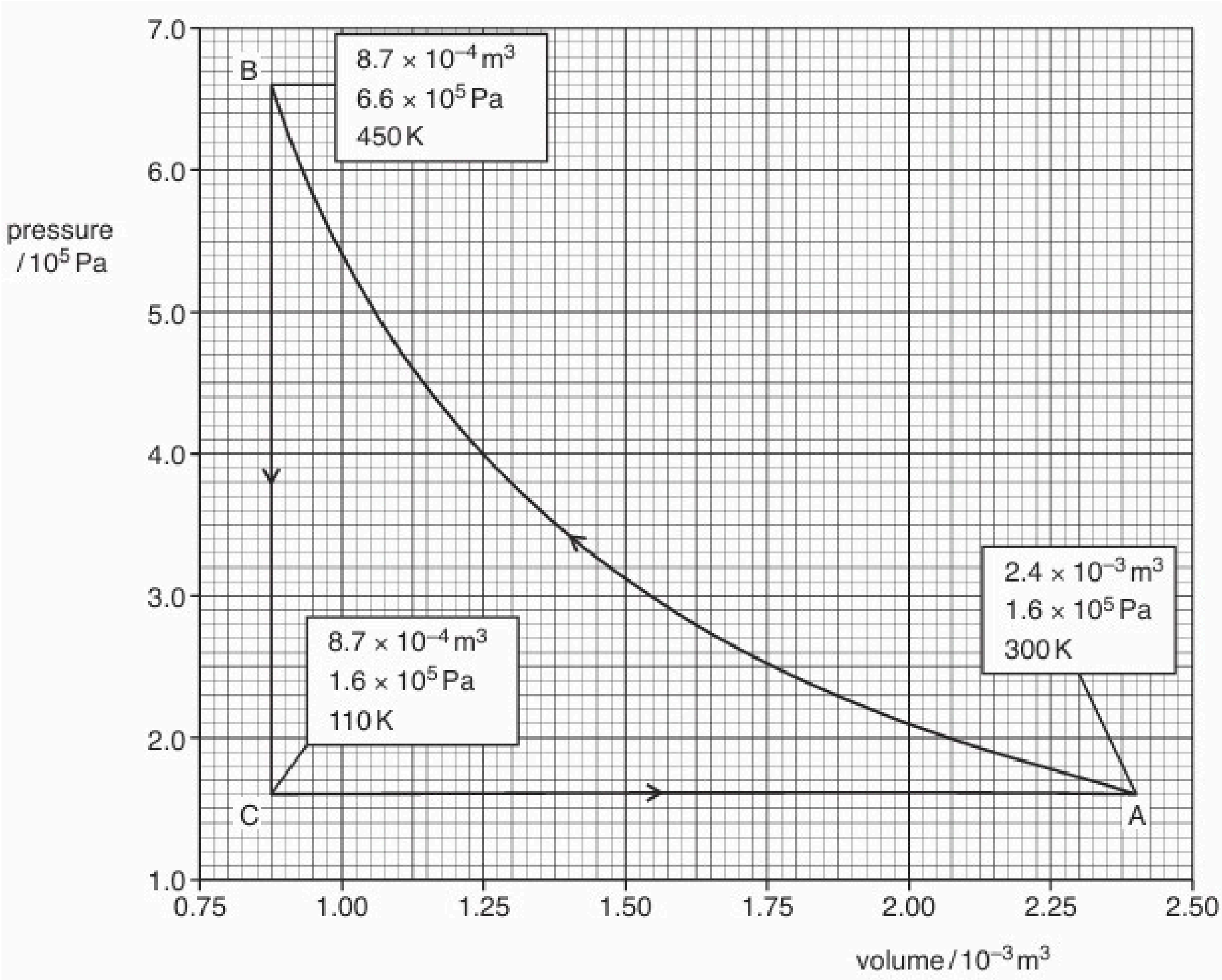


**Fig. 2.1**

At point A, the gas has volume $2.4 \times 10^{-3}$ m$^3$, pressure $1.6 \times 10^5$ Pa and temperature 300 K.

The gas is compressed suddenly so that no thermal energy enters or leaves the gas during the compression. The amount of work done is 480 J so that, at point B, the gas has volume $8.7 \times 10^{-4}$ m$^3$, pressure $6.6 \times 10^5$ Pa and temperature 450 K.

The gas is now cooled at constant volume so that, between points B and C, 1100 J of thermal energy is transferred. At point C, the gas has pressure $1.6 \times 10^5$ Pa and temperature 110 K.

Finally, the gas is returned to point A.

**(i)** State and explain the total change in internal energy of the gas for one complete cycle ABCA.

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.......................................................................................................................................[2]

**(ii)** Calculate the external work done on the gas during the expansion from point C to point A.

work done = ........................................................ J [2]

**(iii)** Complete Fig. 2.2 for the changes from:

**1.** point A to point B

**2.** point B to point C

**3.** point C to point A.

| change | $+q$/J | $+w$/J | $\Delta U$/J |
|---|---|---|---|
| A $\rightarrow$ B | .................... | .................... | .................... |
| B $\rightarrow$ C | .................... | .................... | .................... |
| C $\rightarrow$ A | .................... | .................... | .................... |

**Fig. 2.2**

[4]

[Total: 11]

**16. M/J/20/41/Q2**

**(a)** State what is meant by the *internal energy* of a system.

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.................................................................................................................................. [2]

**(b)** By reference to intermolecular forces, explain why the change in internal energy of an ideal gas is equal to the change in total kinetic energy of its molecules.

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**(c)** State and explain the change, if any, in the internal energy of a solid metal ball as it falls under gravity in a vacuum.

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[Total: 7]

**17. M/J/20/42/Q3**

By reference to the first law of thermodynamics, state and explain the change, if any, in the internal energy of:

**(a)** a lump of solid lead as it melts at constant temperature

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**(b)** some gas in a toy balloon when the balloon bursts and no thermal energy enters or leaves the gas.

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........................................................................................................................................ [3]

[Total: 6]

**18. O/N/20/41/Q2**

**(a)** The first law of thermodynamics may be expressed as

$$\Delta U = (+q) + (+w)$$

where $\Delta U$ is the increase in internal energy of the system.

State the meaning of:

$+q$ ..........................................................................................................................

..........................................................................................................................

$+w$. ..........................................................................................................................

..........................................................................................................................
[2]

**(b)** The variation with pressure $p$ of the volume $V$ of a fixed mass of an ideal gas is shown in Fig. 2.1.

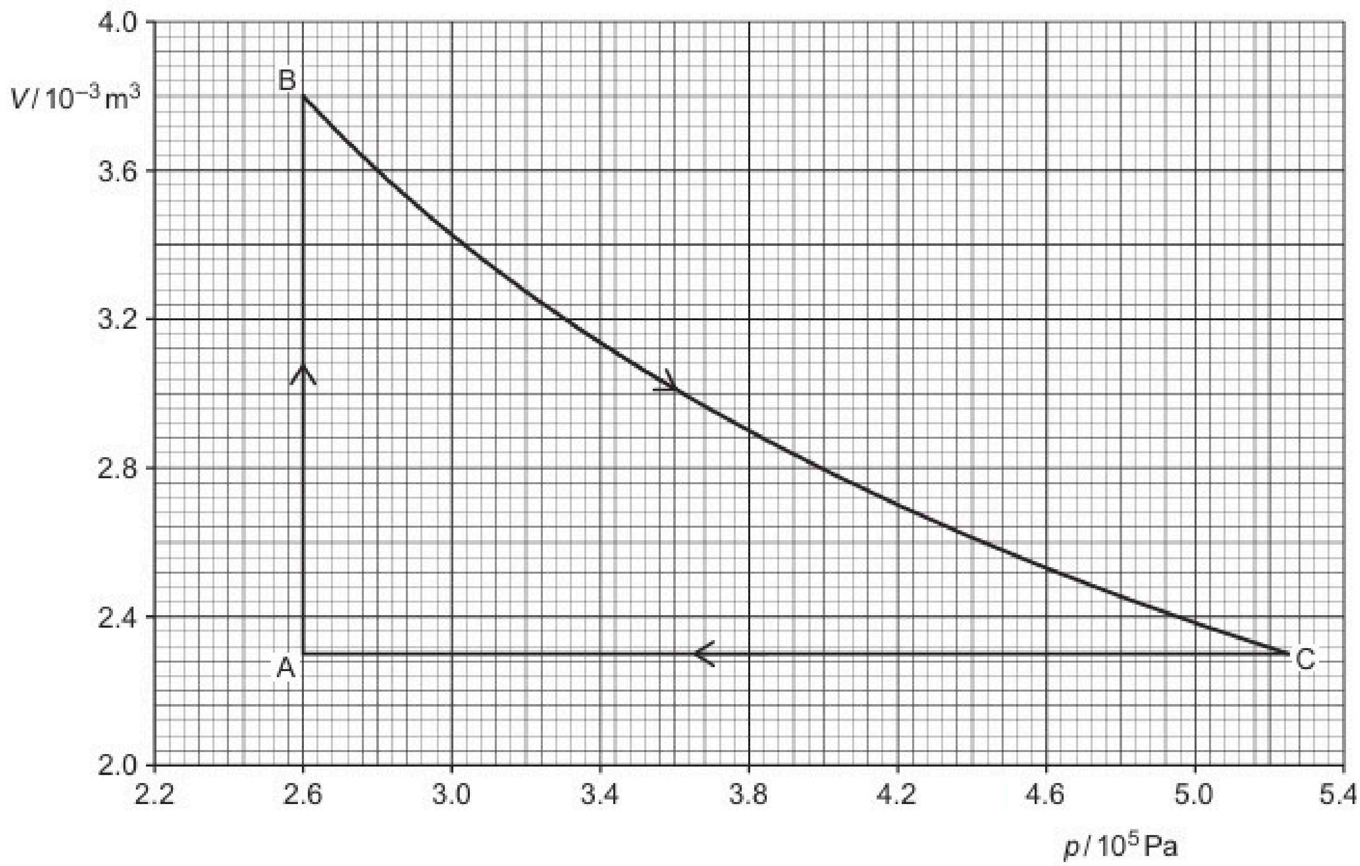


**Fig. 2.1**

The gas undergoes a cycle of changes A to B to C to A.

During the change A to B, the volume of the gas increases from $2.3 \times 10^{-3}\,m^3$ to $3.8 \times 10^{-3}\,m^3$.

(i) Show that the magnitude of the work done during the change A to B is 390 J.

[1]

(ii) State and explain the total change, if any, in the internal energy of the gas during one complete cycle.

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.................................................................................................................................... [2]

**(c)** During the change A to B, 1370 J of thermal energy is transferred to the gas.

During the change B to C, no thermal energy enters or leaves the gas. The work done on the gas during this change is 550 J.

Use these data and the information in **(b)** to complete Table 2.1.

**Table 2.1**

| change | $q$ / J | $w$ / J | $\Delta U$ / J |
|---|---|---|---|
| A to B | .......................... | .......................... | .......................... |
| B to C | .......................... | .......................... | .......................... |
| C to A | .......................... | .......................... | .......................... |

[4]

[Total: 9]

**19. O/N/21/41/Q3**

**(a)** Define *specific heat capacity*.

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.................................................................................................................................... [2]

**(b)** A sealed container of fixed volume $V$ contains $N$ molecules, each of mass $m$, of an ideal gas at pressure $p$.

**(i)** State an expression, in terms of $V$, $N$, $p$ and the Boltzmann constant $k$, for the thermodynamic temperature $T$ of the gas.

.................................................................................................................................... [1]

**(ii)** Show that the mean translational kinetic energy $E_K$ of a molecule of the gas is given by

$$E_K = \frac{3}{2}kT.$$

[2]

**(iii)** Explain why the internal energy of the gas is equal to the total kinetic energy of the molecules.

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.................................................................................................................................... [2]

**(c)** The gas in **(b)** is supplied with thermal energy $Q$.

**(i)** Explain, with reference to the first law of thermodynamics, why the increase in internal energy of the gas is $Q$.

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.................................................................................................................................... [2]

(ii) Use the expression in **(b)(ii)** and the information in **(c)(i)** to show that the specific heat capacity $c$ of the gas is given by

$$c = \frac{3k}{2m}.$$

[2]

**(d)** The container in **(b)** is now replaced with one that does not have a fixed volume. Instead, the gas is able to expand, so that the pressure of the gas remains constant as thermal energy is supplied.

Suggest, with a reason, how the specific heat capacity of the gas would now compare with the value in **(c)(ii)**.

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[Total: 13]

**20. F/M/22/42/Q2**

A fixed mass of an ideal gas has a volume $V$ and a pressure $p$.

The gas undergoes a cycle of changes, X to Y to Z to X, as shown in Fig. 2.1.

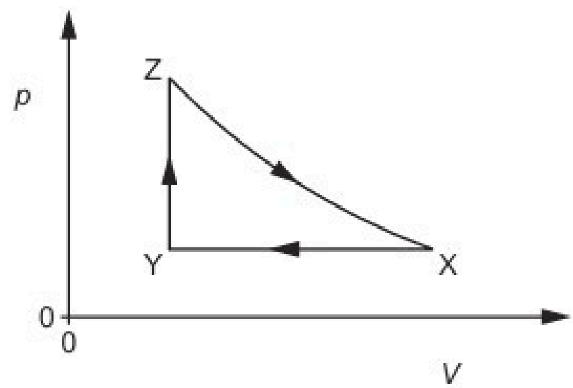


**Fig. 2.1**

Table 2.1 shows data for $p$, $V$ and temperature $T$ for the gas at points X, Y and Z.

**Table 2.1**

| | $p / 10^5$ Pa | $V / 10^{-3}$ m$^3$ | $T$ / K |
|---|---|---|---|
| X | 1.5 | 4.2 | 540 |
| Y | | | 230 |
| Z | 5.1 | | 782 |

**(a)** State the change in internal energy $\Delta U$ for one complete cycle, XYZX.

$\Delta U$ = ........................................................ J [1]

**(b)** Calculate the amount $n$ of gas.

$n$ = .................................................... mol [2]

**(c)** Complete Table 2.1.
Use the space below for any working.

[2]

**(d) (i)** The first law of thermodynamics for a system may be represented by the equation

$$\Delta U = q + W.$$

State, with reference to the system, what is meant by:

$\Delta U$: ..........................................................................................................................................

$q$: ..........................................................................................................................................

$W$: ..........................................................................................................................................
[3]

**(ii)** Explain how the first law of thermodynamics applies to the change Z to X.

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.......................................................................................................................................... [2]

[Total: 10]

**21. M/J/22/41/Q3**

A fixed mass of an ideal gas is initially at a temperature of 17 °C.
The gas has a volume of $0.24\,m^3$ and a pressure of $1.2 \times 10^5\,Pa$.

**(a)** **(i)** State what is meant by an ideal gas.

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.................................................................................................................................... [2]

**(ii)** Calculate the amount $n$ of gas.

$n$ = ................................................... mol [2]

**(b)** The gas undergoes three successive changes, as shown in Fig. 3.1.

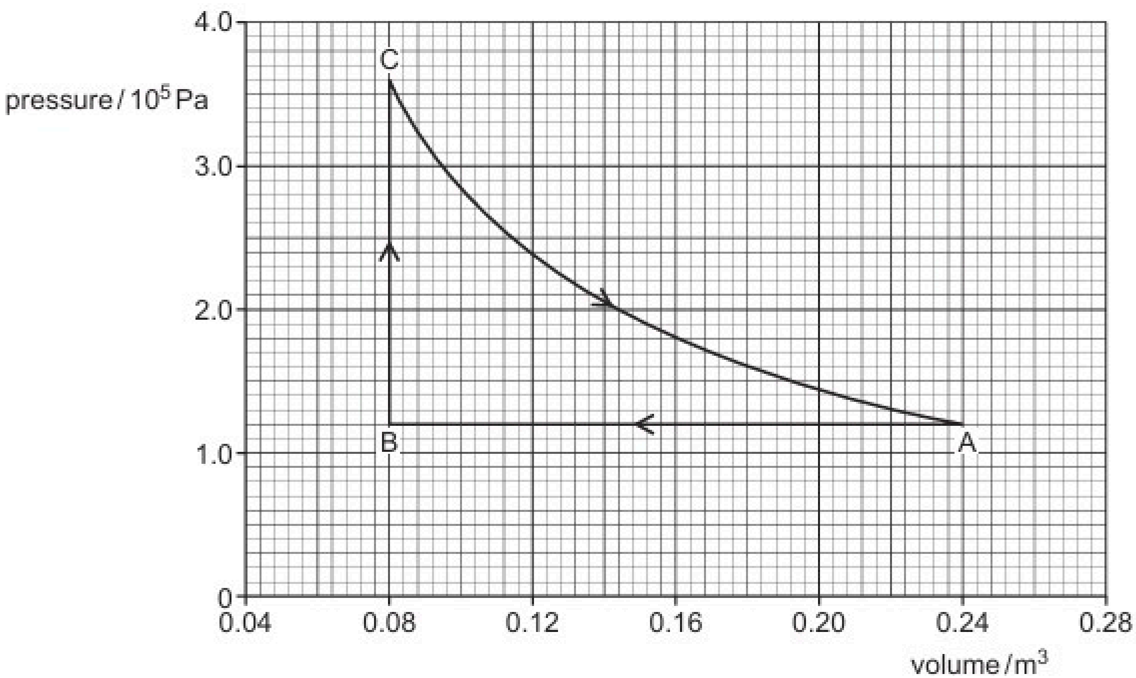


**Fig. 3.1**

The initial state is represented by point A. The gas is cooled at constant pressure to point B by the removal of 48.0 kJ of thermal energy.

The gas is then heated at constant volume to point C.

Finally, the gas expands at constant temperature back to its original pressure and volume at point A. During this expansion, the gas does 31.6 kJ of work.

**(i)** Show that the magnitude of the work done during the change AB is 19.2 kJ.

[2]

**(ii)** Complete Table 3.1 to show the work done on the gas, the thermal energy supplied to the gas and the increase in internal energy of the gas, for each of the changes AB, BC and CA.

**Table 3.1**

| change | work done on gas / kJ | thermal energy supplied to gas / kJ | increase in internal energy of gas / kJ |
|---|---|---|---|
| AB | | −48.0 | |
| BC | | | |
| CA | −31.6 | | |

[5]

[Total: 11]

**22. O/N/22/42/Q2**

**(a)** Define specific heat capacity.

..................................................................................................................................................

..................................................................................................................................................

.......................................................................................................................................... [2]

**(b)** A fixed mass of water in a beaker is at atmospheric pressure.

**(i)** The initial temperature of the water is 0 °C.

The water is supplied with thermal energy $E$, so that its temperature increases to 8 °C. There is no net change in the volume of the water.

Use the first law of thermodynamics to complete Table 2.1 for this process.

**Table 2.1**

| work done on water | thermal energy supplied to water | increase in internal energy of water |
|---|---|---|
| | + $E$ | |

[2]

**(ii)** The water is now heated so that its temperature increases by a further 8 °C to a final temperature of 16 °C. This process causes the volume of the water to increase so that work $W$ is done.
Assume that the change in internal energy is the same as in **(b)(i)**.

Use the first law of thermodynamics to complete Table 2.2 for this process.

**Table 2.2**

| work done on water | thermal energy supplied to water | increase in internal energy of water |
|---|---|---|
| | | |

[2]

**(c)** Use the information in **(b)** to suggest, with a reason, how the average specific heat capacity of water between 8 °C and 16 °C compares with its average value between 0 °C and 8 °C.

..................................................................................................................................................

.......................................................................................................................................... [1]

[Total: 7]

**23. F/M/23/42/Q2**

**(a)** State what is meant by an ideal gas.

..........................................................................................................................................

..........................................................................................................................................

.......................................................................................................................................... [2]

**(b)** A fixed amount of helium gas is sealed in a container. The helium gas has a pressure of $1.10 \times 10^5$ Pa, and a volume of $540\,\text{cm}^3$ at a temperature of 27 °C.

The volume of the container is rapidly decreased to $30.0\,\text{cm}^3$. The pressure of the helium gas increases to $6.70 \times 10^6$ Pa and its temperature increases to 742 °C, as illustrated in Fig. 2.1.

| initial state | final state |
|---|---|
| $1.10 \times 10^5$ Pa | $6.70 \times 10^6$ Pa |
| $540\,\text{cm}^3$ | $30.0\,\text{cm}^3$ |
| 27 °C | 742 °C |

**Fig. 2.1**

No thermal energy enters or leaves the helium gas during this process.

**(i)** Show that the helium gas behaves as an ideal gas.

[2]

**(ii)** The first law of thermodynamics may be expressed as

$$\Delta U = q + W.$$

Use the first law of thermodynamics to explain why the temperature of the helium gas increases.

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

.......................................................................................................................................... [2]

**(iii)** The average translational kinetic energy $E_K$ of a molecule of an ideal gas is given by

$$E_K = \frac{3}{2}kT$$

where $k$ is the Boltzmann constant and $T$ is the thermodynamic temperature.

Calculate the change in the total kinetic energy of the molecules of the helium gas.

change in kinetic energy = ........................................................ J [3]

**(c)** The mass of nitrogen gas in another container is 24.0 g at a temperature of 27 °C. The gas is cooled to its boiling point of –196 °C. Assume all the gas condenses to a liquid.

For this change the specific heat capacity of nitrogen gas is 1.04 $kJ\,kg^{-1}\,K^{-1}$.

The specific latent heat of vaporisation of nitrogen is 199 $kJ\,kg^{-1}$.

Determine the thermal energy, in kJ, removed from the nitrogen gas.

energy = ...................................................... kJ [3]

[Total: 12]

**24. M/J/23/41/Q4**

**(a)** State **two** of the basic assumptions of the kinetic theory of gases.

1 ..............................................................................................................................

..............................................................................................................................

2 ..............................................................................................................................

..............................................................................................................................

[2]

**(b)** An ideal gas has amount of substance $n$.
The gas is initially in state X, with pressure $2p$ and volume $V$.
The gas is cooled at constant volume to state Y, with pressure $p$.
The gas is then heated at constant pressure to state Z, with volume $2V$.
Finally, the gas returns at constant temperature to state X.

**(i)** Determine an expression for the temperature $T$ of the gas in state X, in terms of $n$, $p$ and $V$.
Identify any other symbols that you use.

[2]

**(ii)** On Fig. 4.1, sketch the variation with volume of pressure for the gas as the gas undergoes the three changes. The state X is labelled. Label states Y and Z.

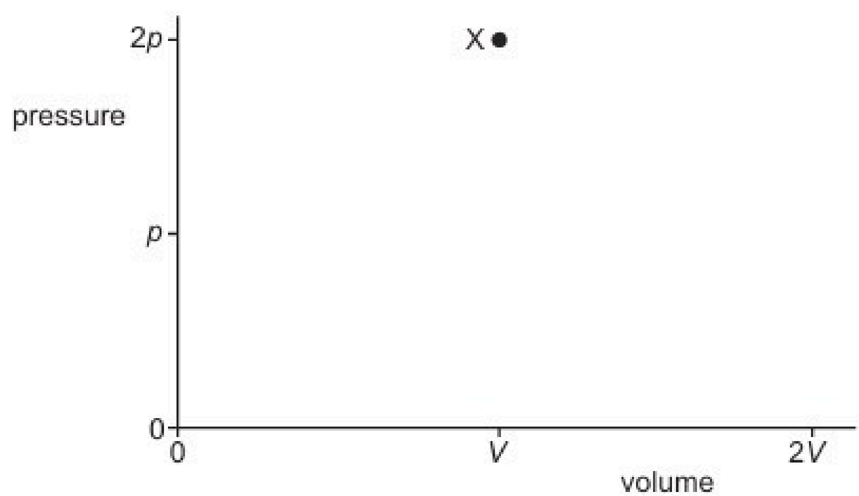


**Fig. 4.1**

[3]

**(iii)** During the change of state from Y to Z, the increase in internal energy of the gas is $U$. During the change of state from Z to X, the work done on the gas is $W$.

Complete Table 4.1 to indicate, for each of the three changes of state, the increase in internal energy of the gas, the thermal energy transferred to the gas and the work done on the gas, in terms of $p$, $V$, $U$ and $W$.

**Table 4.1**

| change | increase in internal energy of gas | thermal energy transferred to gas | work done on gas |
|---|---|---|---|
| X to Y | | | |
| Y to Z | $+U$ | | |
| Z to X | | | $+W$ |

[5]

[Total: 12]

***25. M/J/23/42/Q3***

**(a)** State the first law of thermodynamics. Identify the meaning of any symbols that you use.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [2]

**(b)** The state of an ideal gas is continuously changed according to the cycle ABCDA shown in Fig. 3.1.

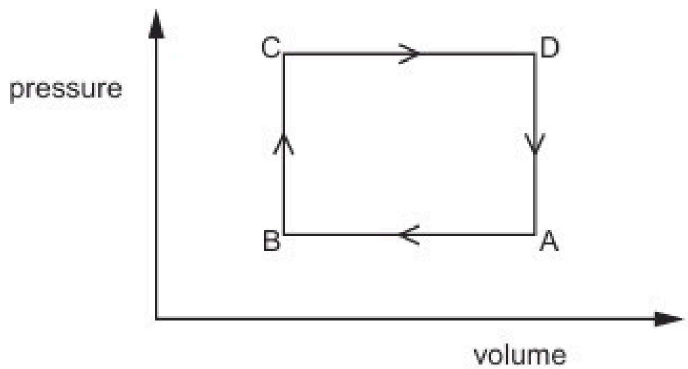


**Fig. 3.1**

**(i)** Complete Table 3.1 for the changes A to B and B to C by placing two ticks (✓) in each row.

**Table 3.1**

| change | change in internal energy | | | work done on gas | | |
|---|---|---|---|---|---|---|
| | decrease | no change | increase | negative | zero | positive |
| A to B | | | | | | |
| B to C | | | | | | |

[4]

**(ii)** Use the first law of thermodynamics to describe and explain the energy transfers associated with one complete cycle ABCDA.

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..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [3]

[Total: 9]

**26. O/N/23/41/Q2**

**(a)** Define specific heat capacity.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [2]

**(b)** An ideal gas of mass 0.35 kg is heated at a constant pressure of $2.0 \times 10^5$ Pa so that its internal energy increases by 7600 J. During this process, the volume of the gas increases from 0.038 $m^3$ to 0.063 $m^3$ and the temperature increases by 56 °C.

**(i)** Show that the magnitude of the work done on the gas is 5000 J.

[1]

**(ii)** Explain whether the work done on the gas is positive or negative.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [2]

**(iii)** Determine the magnitude of the thermal energy $q$ transferred to the gas.

$q$ = ...................................................... J [2]

**(iv)** Calculate the specific heat capacity of the gas for this process. Give a unit with your answer.

specific heat capacity = ............................................ unit .............. [2]

**(c)** The gas in **(b)** is now heated at constant volume rather than at constant pressure. The increase in internal energy of the gas is the same as in **(b)**.

Use the first law of thermodynamics to explain whether the specific heat capacity of the gas for this process is less than, the same as, or greater than the answer in **(b)(iv)**.

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [3]

[Total: 12]

**27. O/N/23/42/Q3**

**(a)** State what is meant by the internal energy of a system.

..........................................................................................................................................

..........................................................................................................................................

.......................................................................................................................................... [2]

**(b)** Use the first law of thermodynamics to explain what happens to the internal energy:

**(i)** of a spring when it is stretched at constant temperature within its elastic limit

..........................................................................................................................................

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..........................................................................................................................................

.......................................................................................................................................... [3]

**(ii)** of a sample of water when it evaporates from a rain puddle on a hot day.

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..........................................................................................................................................

.......................................................................................................................................... [3]

[Total: 8]

# *ANSWERS*

***1.***

**(a)** sum of kinetic and potential energies of molecules / particles / atoms ......................M1
random (distribution) ........................................................................................ A1

**(b)** $+\Delta U$: increase in internal energy ........................................................................ B1
$+q$: heating of / heat supplied to system ................................................................ B1
$+w$: work done on system ........................................................................................ B1

**(c)** **(i)** work done $= p\Delta V$ ........................................................................................C1
$= 1.0 \times 10^5 \times (2.1 - 1.8) \times 10^{-3}$
$= 30$ J ........................................................................................M1
$w = 30$ J, $q = 0$ so $\Delta U = 30$ J ........................................................................ A1

***2.***

**(a)** **(i)** no forces (of attraction or repulsion) between atoms / molecules / particles B1

**(ii)** sum of kinetic and potential energy of atoms / molecules M1
due to random motion A1

**(iii)** (random) kinetic energy increases with temperature M1
no potential energy
(so increase in temperature increases internal energy) A1

**(b)** **(i)** zero A1

**(ii)** work done $= p\Delta V$ C1
$= 4.0 \times 10^5 \times 6 \times 10^{-4}$
$= 240$ J *(ignore any sign)* A1

**(iii)**

| change | work done / J | heating / J | increase in internal energy / J |
|---|---|---|---|
| P → Q | **+240** | –600 | **–360** |
| Q → R | 0 | +720 | **+720** |
| R → P | **–840** | +480 | **–360** |

(*correct signs essential*)
(*each horizontal line correct, 1 mark – max 3*) B3

*3.*

**(a)** $+\Delta U$: increase in internal energy B1

$+q$: thermal energy / heat supplied to the system B1

$+w$: work done on the system B1

**(b) (i)** (thermal) energy required to change the state of a substance M1

per unit mass A1

without any change of temperature A1

**(ii)** when evaporating

greater change in separation of atoms/molecules M1

greater change in volume M1

identifies each difference correctly with $\Delta U$ and $w$ A1

*4.*

**(a)** sum of potential energy and kinetic energy of atoms/molecules/particles M1

reference to random (distribution) A1

**(b) (i)** as lattice structure is 'broken'/bonds broken/forces between molecules reduced (not molecules separate) B1

no change in kinetic energy, potential energy increases M1

internal energy increases A1

**(ii)** *either* molecules/atoms/particles move faster/ $<c^2>$ is increasing

*or* kinetic energy increases with temperature (increases) B1

no change in potential energy, kinetic energy increases M1

internal energy increases A1

*5.*

**(a) (i)** sum of potential energy and kinetic energy of atoms / molecules / particles M1

reference to random A1

**(ii)** no intermolecular forces B1

no potential energy B1

internal energy is kinetic energy (of random motion) of molecules B1

(*reference to random motion here then allow back credit to **(i)** if M1 scored*)

**(b)** kinetic energy $\propto$ thermodynamic temperature B1

*either* temperature in Celsius, not kelvin so incorrect

*or* temperature in kelvin is not doubled B1

*6.*

| | | | |
|---|---|---|---|
| **(a)** | **(i)** | sum of kinetic and potential energies of the molecules<br>reference to random distribution | M1<br>A1 |
| | **(ii)** | for ideal gas, no intermolecular forces<br>so no potential energy (only kinetic) | M1<br>A1 |
| **(b)** | **(i)** | *either* change in kinetic energy $= 3/2 \times 1.38 \times 10^{-23} \times 1.0 \times 6.02 \times 10^{23} \times 180$<br>$= 2240\,\text{J}$<br>*or* $R = kN_A$<br>energy $= 3/2 \times 1.0 \times 8.31 \times 180$<br>$= 2240\ \text{J}$ | C1<br>A1<br><br>(C1)<br>(A1) |
| | **(ii)** | increase in internal energy = heat supplied + work done on system<br>2240 = energy supplied – 1500<br>energy supplied = 3740 J | B1<br>C1<br>A1 |

*7.*

| | | | |
|---|---|---|---|
| **(a)** | | *either* change in volume $= (1.69 - 1.00 \times 10^{-3})$<br>*or* liquid volume $<<$ volume of vapour<br>work done $= 1.01 \times 10^5 \times 1.69 = 1.71 \times 10^5$ (J) | <br>M1<br>A1 |
| **(b)** | **(i)** | **1.** heating of system/thermal energy supplied to the system | B1 |
| | | **2.** work done on the system | B1 |
| | **(ii)** | $\Delta U = (2.26 \times 10^6) - (1.71 \times 10^5)$<br>$= 2.09 \times 10^6$ J (*3 s.f. needed*) | C1<br>A1 |

*8.*

| | | | |
|---|---|---|---|
| **(a)** | | (sum of) potential energy and kinetic energy of molecules/atoms/particles<br>mention of random motion/distribution | M1<br>A1 |
| **(b)** | **(i)** | $pV = nRT$<br>*either* at A, $1.2 \times 10^5 \times 4.0 \times 10^{-3} = n \times 8.31 \times 290$<br>*or* at B, $3.6 \times 10^5 \times 4.0 \times 10^{-3} = n \times 8.31 \times 870$<br>$n = 0.20$ mol | <br><br>C1<br>A1 |
| | **(ii)** | $1.2 \times 10^5 \times 7.75 \times 10^{-3} = 0.20 \times 8.31 \times T$ *or* $T = (7.75/4.0) \times 290$<br>$T = 560$ K<br>(*Allow tolerance from graph*: $7.7–7.8 \times 10^{-3}\,\text{m}^3$) | C1<br>A1 |
| **(c)** | | temperature changes/decreases so internal energy changes/decreases<br>volume changes (at constant pressure) so work is done | B1<br>B1 |

*9.*

**(a)** heat/thermal energy gained by system *or* energy transferred to system by heating B1
plus work done on the system *or* minus work done by the system B1

**(b)** **(i)** *either* volume decreases so work done on the system
*or* small volume change so work done on system negligible M1
(thermal) energy absorbed to break lattice structure M1
internal energy increases A1

**(ii)** gas expands so work done by gas (against atmosphere) M1
no time for thermal energy to enter or leave the gas M1
internal energy decreases A1

*10.*

**(a)** **(i)** $p \propto T$ *or* $pV/T$ = constant *or* $pV = nRT$ C1

$T$ (= 5 × 300 =) 1500 K A1

**(ii)** $pV = nRT$

$1.0 \times 10^5 \times 4.0 \times 10^{-4} = n \times 8.31 \times 300$
*or*
$5.0 \times 10^5 \times 4.0 \times 10^{-4} = n \times 8.31 \times 1500$ C1

$n$ = 0.016 mol A1

**(b)** **(i)** **1.** heating/thermal energy supplied B1

**2.** work done on/to system B1

**(ii)** **1.** 240 J A1

**2.** same value as given in **1.** (= 240 J) **and** zero given for **3.** A1

**3.** zero A1

*11.*

**(a)** (sum of/total) potential energy and kinetic energy of (all) molecules/particles M1
reference to random (distribution) A1

**(b)** **(i)** no heat enters (gas)/leaves (gas)/no heating (of gas) B1

work done by gas (against atmosphere as it expands) M1

internal energy decreases A1

**(ii)** volume decreases so work done on ice/water (*allow work done negligible because $\Delta V$ small*) B1

heating of ice (to break rigid forces/bonds) M1

internal energy increases A1

***12.***

| | | |
|---|---|---|
| 2(a) | $+\Delta U$ <u>increase</u> in internal energy<br>$+q$ heat (energy) transferred <u>to</u> the system / heating of system<br>$+w$ work done <u>on</u> system | **B2** |
| 2(b)(i) | $W = p\Delta V$<br>$= 5.2 \times 10^5 \times (5.0 - 1.6) \times 10^{-4}$ (=177 J) | **B1** |
| | $\Delta U = q + w$<br>$= 442 - 177 = 265$ J | **A1** |
| 2(b)(ii) | no (molecular) potential energy | **B1** |
| | internal energy decreases so (total molecular) kinetic energy decreases | **B1** |
| | (mean molecular) kinetic energy decreases so temperature decreases | **B1** |
| 2(b)(iii) | $\Delta U + 265 - 313 = 0$<br>$\Delta U = 48$ J | **A1** |
| 2(b)(iv) | $pV = NkT$ or $pV = nRT$ <u>and</u> $N = nN_A$ | **C1** |
| | $5.2 \times 10^5 \times 1.6 \times 10^{-4} = N \times 1.38 \times 10^{-23} \times (273 + 227)$<br>*or*<br>$5.2 \times 10^5 \times 1.6 \times 10^{-4} = n \times 8.31 \times (273 + 227)$ and $n = N / 6.02 \times 10^{23}$<br><br>$N = 1.2 \times 10^{22}$ | **A1** |

***13.***

| | | |
|---|---|---|
| 3(a)(i) | sum of potential and kinetic energies (of molecules/atoms/particles) | **B1** |
| | (energy of) molecules/atoms/particles in random motion | **B1** |
| 3(a)(ii) | (in ideal gas) no intermolecular forces so no potential energy | **B1** |
| | internal energy is (solely) kinetic energy (of particles) | **B1** |
| | (mean) kinetic energy (of particles) proportional to (thermodynamic) temperature of gas | **B1** |
| 3(b) | $pV = NkT$ | **C1** |
| | $6.4 \times 10^6 \times 1.8 \times 10^4 \times 10^{-6} = N \times 1.38 \times 10^{-23} \times 298$ | **C1** |
| | **or** | |
| | $pV = nRT$ **and** $N = n \times N_A$ | **(C1)** |
| | $6.4 \times 10^6 \times 1.8 \times 10^4 \times 10^{-6} = n \times 8.31 \times 298$<br><br>$n = 46.5$ (mol)<br><br>$N = 46.5 \times 6.02 \times 10^{23}$ | **(C1)** |
| | $N = 2.8 \times 10^{25}$ | **A1** |

## 14.

| | | |
|---|---|---|
| 2(a) | sum of potential and kinetic energies (of molecules/atoms/particles) | **B1** |
| | (energy of) molecules/atoms/particles in random motion | **B1** |
| 2(b)(i) | final temperature = initial temperature | **B1** |
| | no change in internal energy | **B1** |
| 2(b)(ii) | **1.** work done on gas (P→Q): 0 | **A1** |
| | increase in internal energy (P→Q): (+)97.0 J | **A1** |
| | **2.** increase in internal energy (Q→R): –42.5 J | **A1** |
| | **3.** increase in internal energy (R→P): –54.5 J | **A1** |
| | thermal energy supplied (R→P): –91.5 J | **A1** |

## 15.

| | | |
|---|---|---|
| 2(a)(i) | **1.** energy transfer <u>to</u> the system by heating | **B1** |
| | **2.** (external) work done <u>on</u> the system | **B1** |
| 2(a)(ii) | <u>decrease</u> in internal energy | **B1** |
| 2(b)(i) | no change (in internal energy) | **B1** |
| | (because) no change in temperature | **B1** |
| 2(b)(ii) | work done $= p\Delta V$<br>$= (–)1.6 \times 10^{5} \times (2.4 – 0.87) \times 10^{-3}$ | **C1** |
| | $= (–)240$ J | **A1** |
| 2(b)(iii) | first row all correct (0, 480, 480) | **A1** |
| | second row all correct (–1100, 0, –1100) | **A1** |
| | final column of third row calculated correctly from the two values above it, so that the final column adds up to 0 | **A1** |
| | second column in final row correct, with correct negative sign<br>**and**<br>first column in final row calculated correctly so that it adds to the second column to give the third column<br><br>(fully correct table is:<br><br>0 \| 480 \| 480<br>–1100 \| 0 \| –1100<br>860 \| –240 \| 620<br>) | **A1** |

## 16.

| | | |
|---|---|---|
| 2(a) | total potential energy and kinetic energy (of molecules/atoms) | **M1** |
| | reference to <u>random</u> motion of molecules/atoms | **A1** |
| 2(b) | (in ideal gas,) no intermolecular forces | **B1** |
| | no potential energy (so change in kinetic energy is change in internal energy) | **B1** |

| | | |
|---|---|---|
| 2(c) | (random) potential energy of molecules does not change | **M1** |
| | (random) kinetic energy of molecules does not change | **M1** |
| | so internal energy does not change | **A1** |
| | **or** | |
| | decrease in total potential energy = gain in total kinetic energy | **(M1)** |
| | no external energy supplied | **(M1)** |
| | so internal energy does not change | **(A1)** |
| | **or** | |
| | no compression (of ball) so no work done on the ball | **(M1)** |
| | no resistive forces so no heating of the ball | **(M1)** |
| | so internal energy does not change | **(A1)** |

***17.***

| | | |
|---|---|---|
| 3(a) | (little/no volume change so) little/no external work done | **B1** |
| | thermal energy supplied to provide latent heat | **M1** |
| | internal energy increases | **A1** |
| 3(b) | (rapid) increase in volume | **B1** |
| | gas does work against the atmosphere | **M1** |
| | internal energy decreases | **A1** |

***18.***

| | | |
|---|---|---|
| 2(a) | $+q$: thermal energy transfer to system | **B1** |
| | $+w$: work done on system | **B1** |
| 2(b)(i) | ($W$ =) $2.6 \times 10^5 \times (3.8 - 2.3) \times 10^{-3} = 390$ J | **A1** |
| 2(b)(ii) | no (total) change (in internal energy) | **B1** |
| | gas returns to its original temperature | **B1** |
| 2(c) | A to B row all correct (1370, – 390, 980) | **B1** |
| | B to C row all correct (0, 550, 550) | **B1** |
| | C to A row: $\Delta U$ adds to the other two $\Delta U$ values to give zero | **B1** |
| | C to A row: $w = 0$ **and** $q$ adds to $w$ to give $\Delta U$ value<br>complete correct answer: | **B1** |

| change | $q$ / J | $w$ / J | $\Delta U$ / J |
|---|---|---|---|
| A to B | (+)1370 | –390 | (+)980 |
| B to C | 0 | (+)550 | (+)550 |
| C to A | –1530 | 0 | –1530 |

## 19.

| | | |
|---|---|---|
| 3(a) | (thermal) energy per unit mass (to cause temperature change) | **B1** |
| | (thermal) energy per unit change in temperature | **B1** |
| 3(b)(i) | ($T$ =) $pV / Nk$ | **B1** |
| 3(b)(ii) | ($pV$ =) $NkT = ⅓Nm\langle c^2\rangle$ **or** $pV = NkT$ **and** $pV = ⅓Nm\langle c^2\rangle$ | **M1** |
| | leading to $½m\langle c^2\rangle = (3/2)kT$ and $½m\langle c^2\rangle = E_K$ | **A1** |
| 3(b)(iii) | internal energy = $\Sigma E_K$ (of molecules) + $\Sigma E_P$ (of molecules) **or** no forces between molecules | **B1** |
| | potential energy of molecules is zero | **B1** |
| 3(c)(i) | increase in internal energy = $Q$ + work done | **B1** |
| | constant volume so no work done | **B1** |
| 3(c)(ii) | $c = Q / Nm\Delta T$ | **C1** |
| | $= [N \times (3/2)k\Delta T] / (Nm\Delta T) = 3k / 2m$ | **A1** |
| 3(d) | (as it expands) gas does work (against the atmosphere/external pressure) | **B1** |
| | for same temperature rise) more (thermal) energy needed, so larger specific heat capacity | **B1** |

## 20.

| | | |
|---|---|---|
| 2(a) | 0 | **B1** |
| 2(b) | $pV = nRT$<br>($n$ =) $1.5 \times 10^5 \times 4.2 \times 10^{-3} / 8.31 \times 540$ | **C1** |
| | = **0.14** mol | **A1** |
| 2(c) | missing pressure **1.5** ($\times 10^5$) | **B1** |
| | both missing volumes **1.8** ($\times 10^{-3}$) | **B1** |
| 2(d)(i) | ($\Delta U$:) increase in internal energy (of the system) | **B1** |
| | ($q$:) thermal energy supplied to the system | **B1** |
| | ($W$:) work done on system | **B1** |
| 2(d)(ii) | volume increases and work is done by the gas | **B1** |
| | temperature decreases and internal energy decreases | **B1** |

## 21.

| | | |
|---|---|---|
| 3(a)(i) | a gas that obeys $pV \propto T$ | M1 |
| | where $p$ = pressure, $V$ = volume, $T$ = thermodynamic temperature | A1 |
| 3(a)(ii) | $T = (273 + 17)$ K | C1 |
| | $n = pV / RT$<br>$= (1.2 \times 10^5 \times 0.24) / [8.31 \times (273 + 17)]$<br>$= 12$ mol | A1 |
| 3(b)(i) | work done $= p\Delta V$ | C1 |
| | $= 1.2 \times 10^5 \times (0.24 - 0.08) = 19200$ J (= 19.2 kJ) | A1 |
| 3(b)(ii) | AB work done correct (19.2) | A1 |
| | BC work done correct (0) | A1 |
| | CA increase in internal energy correct (0) **and** CA thermal energy correct (31.6) | A1 |
| | AB increase in internal energy calculated correctly from work done – 48.0 | A1 |
| | BC increase in internal energy correctly calculated so the final column adds up to zero **and** BC thermal energy same as increase in internal energy<br><br>(Fully correct table:<br>AB \| 19.2 \| –48.0 \| –28.8<br>BC \| 0 \| 28.8 \| 28.8<br>CA \| –31.6 \| 31.6 \| 0<br>) | A1 |

## 22.

| | | |
|---|---|---|
| 2(a) | (thermal) energy per unit mass (to cause temperature change) | B1 |
| | (thermal) energy per unit change in temperature | B1 |
| 2(b)(i) | work done correct (0) | B1 |
| | increase in internal energy correct (+$E$) | B1 |
| 2(b)(ii) | work done correct (–$W$) **and** increase in internal energy same as **(b)(i)** | B1 |
| | thermal energy correct so that it adds to work done to give increase in internal energy | B1 |
| 2(c) | more thermal energy needed so specific heat capacity is greater | B1 |

## 23.

| | | |
|---|---|---|
| 2(a) | gas for which $pV \propto T$ | M1 |
| | where $T$ is thermodynamic temperature | A1 |
| 2(b)(i) | evidence of two temperature conversions between °C and K | B1 |
| | two calculations shown, one for each state e.g.<br>$\frac{1.10 \times 10^5 \times 540 \times 10^{-6}}{(273+27)} = 0.198$ **and** $\frac{6.70 \times 10^6 \times 30 \times 10^{-6}}{(273+742)} = 0.198$ | A1 |
| 2(b)(ii) | work is done on the gas | M1 |
| | internal energy increases (so temperature increases) | A1 |

| | | |
|---|---|---|
| 2(b)(iii) | $pV = NkT$ e.g. $N = \dfrac{1.10 \times 10^{5} \times 540 \times 10^{-6}}{1.38 \times 10^{-23} \times 300}$ $= 1.435 \times 10^{22}$ $\Delta E_k = (3/2)\, k \Delta T N$ | C1 |
| | $= (3/2) \times 1.38 \times 10^{23} \times (742 - 27) \times \dfrac{1.10 \times 10^{5} \times 540 \times 10^{-6}}{1.38 \times 10^{-23} \times 300}$ | C1 |
| | $= 212$ J | A1 |
| 2(c) | $E = mc\Delta\theta$ **and** $E = mL$ | C1 |
| | $\Delta\theta = (27 + 196)$ or 223 | C1 |
| | $E = 0.0240 \times 1.04 \times (27 + 196) + 0.0240 \times 199$ $= 10.3$ kJ | A1 |

## *24.*

| | | |
|---|---|---|
| 4(a) | • particles are in (continuous) random motion<br>• particles have negligible volume (compared with the gas)<br>• negligible forces between particles (except during collisions)<br>• (all) collisions (perfectly) elastic<br>• time of collision negligible (in comparison with time between collisions)<br>*Any two points, 1 mark each* | B2 |
| 4(b)(i) | (general starting equation) $pV = nRT$ | C1 |
| | $T = (2pV / nR)$ where $R$ is the (molar) gas constant | A1 |
| 4(b)(ii) | sketch: straight vertical line XY from ($V$, 2$p$) to ($V$, $p$) | B1 |
| | straight horizontal line YZ from ($V$, $p$) to (2$V$, $p$) | B1 |
| | curve with gradient increasing from Z to X from (2$V$, $p$) to ($V$, 2$p$) | B1 |
| 4(b)(iii) | XY work done on gas correct (= 0) | B1 |
| | ZX increase in internal energy correct (= 0) | B1 |
| | YZ work done on gas correct (= $-pV$) | B1 |
| | XY increase in internal energy such that the increase in internal energy column adds up to zero | B1 |
| | all three thermal energies transferred such that $\Delta U = q + w$ in each row<br>(completely correct answer: | B1 |

| change | $\Delta U$ | $Q$ | $w$ |
|---|---|---|---|
| X to Y | $-U$ | $-U$ | 0 |
| Y to Z | [ $+U$ ] | $U + pV$ | $-pV$ |
| Z to X | 0 | $-W$ | [ $+W$ ] |

)

## 25.

| | | |
|---|---|---|
| 3(a) | <u>change</u> in internal energy = work done + energy transfer by heating | **C1** |
| | <u>increase</u> in internal energy = work done <u>on</u> system + energy transferred <u>to</u> the system by heating | **A1** |
| 3(b)(i) | AB change in internal energy: decrease | **B1** |
| | AB work done on gas: positive | **B1** |
| | BC change in internal energy: increase | **B1** |
| | BC work done on gas: zero | **B1** |
| 3(b)(ii) | more work done by gas in CD than is done on gas in AB<br>**or**<br>(no work done on gas in BC and DA so) (overall) gas does work | **B1** |
| | (overall) change in internal energy is zero | **B1** |
| | (must be an overall) input of thermal energy | **B1** |

## 26.

| | | |
|---|---|---|
| 2(a) | (thermal) energy per unit mass (to change temperature) | **B1** |
| | (thermal) energy per unit change in temperature | **B1** |
| 2(b)(i) | work done $= p\Delta V$<br>$= (2.0 \times 10^5) \times (0.063 - 0.038) = 5000$ J | **A1** |
| 2(b)(ii) | gas is expanding (against external pressure) | **B1** |
| | gas does work / work is done by gas, so (work done on gas is) negative | **B1** |
| 2(b)(iii) | $\Delta U = q + W$ | **C1** |
| | $7600 = q + (-5000)$<br>$q = 12\,600$ J | **A1** |
| 2(b)(iv) | specific heat capacity $= q / m\Delta T$<br>$= 12600 / (0.35 \times 56)$ | **C1** |
| | $= 640$ J kg$^{-1}$ K$^{-1}$ | **A1** |
| 2(c) | same gain in internal energy so same temperature rise | **B1** |
| | no change in volume so no work done<br>**or**<br>no work done so less thermal energy needed (for same change in internal energy) | **B1** |
| | less thermal energy needed (for same temperature change) so lower specific heat capacity | **B1** |

## 27.

| | | |
|---|---|---|
| 3(a) | sum of potential energy and kinetic energy (of particles) | **B1** |
| | (total) energy of random motion of particles | **B1** |
| 3(b)(i) | no thermal energy <u>transferred</u> | **B1** |
| | work is done on the spring (increasing the potential energy of particles) | **M1** |
| | so internal energy increases | **A1** |
| 3(b)(ii) | thermal energy transferred to water | **B1** |
| | work is done by water (expanding against atmosphere as it vaporises) | **B1** |
| | more thermal energy transferred than work done so internal energy increases | **B1** |